MIL-C-24145/24C(SHIPS)
15 November 1971
SUPERSEDING
MIL-C-24145/24B(SHIPS)
4 October 1971

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, SPECIAL PURPOSE, 600 VOLTS A.C.,

FOR SHIPBOARD USE, TYPE 2WA-40

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-24145(SHIPS)

REQUIREMENTS:

Construction (watertight)

First - Standard copper conductor, A.W.G. No. 22 (7/.010 inch), tin coated, maximum outside diameter 0.033 inch.
Second - Polyethylene insulation type II, grade 7 of L-P-390, 0.020 inch minimum wall, color coded (see telephone cable coding).
Third - Insulation jacket of clear polyamide, type III, grade E of MIL-M-20693, nominal wall thickness 0.004 inch.
Fourth - Two insulated and jacketed conductors twisted together to form a pair. Nominal lay length 1.5 inches.
Fifth - Forty pairs cabled together. Maximum lay length 16 times pitch diameter of the layer. Color code and cabling sequence in accordance with requirements for telephone cable coding. All fillers shall be non-fibrous.
Sixth - A binder tape approved by NAVSEC, Electrical Equipment Branch, applied helically with overlap. (See Note 1).
Seventh - Shielding braid of A.W.G. No. 34, tin coated copper. Minimum coverage 85 percent.
Eighth - A polyester barrier tape.
Ninth - Standard thermoplastic jacket. Nominal wall thickness 0.060 inch.
Tenth - Braided metal armor.
Eleventh - Paint.

NOTES:

1. Since the capacitance of all pairs should be approximately the same, it may be necessary to apply a greater thickness of insulation to conductors in the outer layer, because of their proximity to the overall shield braid. For the same reason, the binder tape should be one having good electrical properties.
2. Overall diameter of completed cable shall be 1.420 inches maximum, and 1.370 inches minimum.

EXAMINATION AND TESTS:

Group A:	Requirements:
Visual and dimensional	
Conductor resistance-ohms/1000 ft. @25°C., maximum	17.7
Dielectric strength-volts r.m.s., minimum	
Conductor to shield	2500
Shield to armor or to water	1000
Insulation resistance-megohms/1000 ft., minimum	
Conductor to conductor	5000
Capacitance-pF/ft. @ 1 MHZ, maximum	
(C) Each conductor	30
Each pair of conductors	20
Watertightness-@ 25 p.s.i., cu. in., 6 hours, maximum	1.0

Group B:	
Cold bend - Mandrel diameter 12 times cable diameter	
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - p.s.i. minimum	1400
Elongation-percent, minimum	300
Jacket (cable)	
Tensile strength, p.s.i., minimum	1800
Elongation, percent, minimum	225

(C) Denotes changes.

FSC 6145

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EXAMINATION AND TESTS:	Requirements:
Group C:	
Physicals (accelerated aging)	
Insulation (conductor)	
Air oven	
Elongation-percent of original, minimum	60
Jacket (cable)	
Hot oil immersion	
Tensile strength - p.s.i., minimum	1400
Elongation - percent, minimum	50
Cable filler removability	

Additional test for qualification only:
Cable aging and compatibility (95°C.) ..... No failures

UNIT ORDERING LENGTH: 1000 feet (Nominal).

Preparing activity:
Navy-SH
(Project 6145-N215)

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